



Book I. Prop. XLVIII. Theor. *If the square of one side (—) of a triangle is equal to the squares of the other two sides (— and —), the angle (◡) subtended by that side is a right angle.*

Draw $\perp$ — and
 = — (pr. I.11, I.3)
 and draw also.

Since = — (const.)
² = —²;

$$\therefore \text{.....}^2 + \text{—}^2 = \text{—}^2 + \text{—}^2$$

but² + —² =
² (pr. I.47),

and

$$\text{—}^2 + \text{—}^2 = \text{—}^2 \text{ (hyp.)}$$

$$\therefore \text{.....}^2 = \text{—}^2,$$

$$\therefore \text{.....} = \text{—};$$

and $\therefore \text{◡} = \text{◡}$ (pr. I.8),

consequently ◡ is a right angle.

Q. E. D.